A Low-Complexity Speech Codec Using Parametric Dithering for ASR

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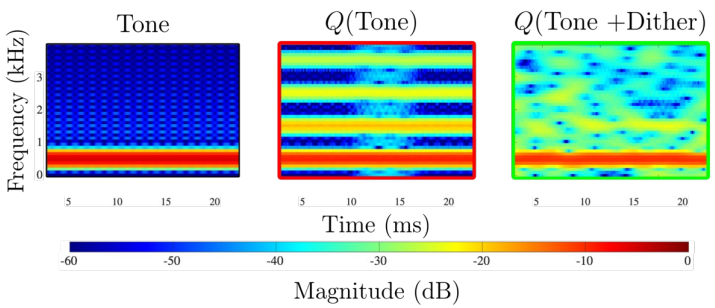
MOTIVATION

- Audio codecs (like MP3 or AAC):
  - optimized for human hearing, not for ML.
  - leave potential redundancies when applied to ML.
- **Low-complexity** waveform quantization introduces significant **harmonic distortion** at low rates.
- **Dithering** prior to quantization Q(.) :
  - ↓ the harmonic distortion.
  - ↑ the additive noise power.
  - ↑ the entropy.



- Automatic Speech Recognition:
  - raw speech audio → text.
  - "sees" audio as a spectrogram
  - sensitive to structural noise in the audio signal.

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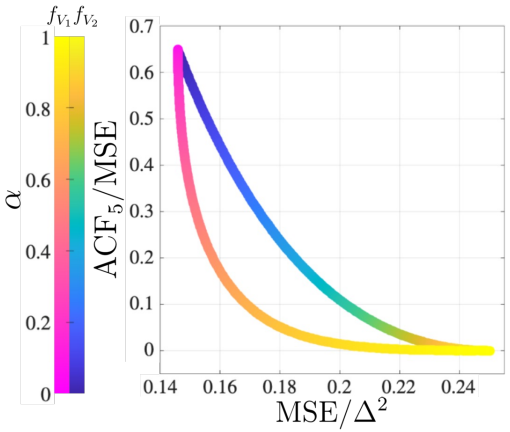
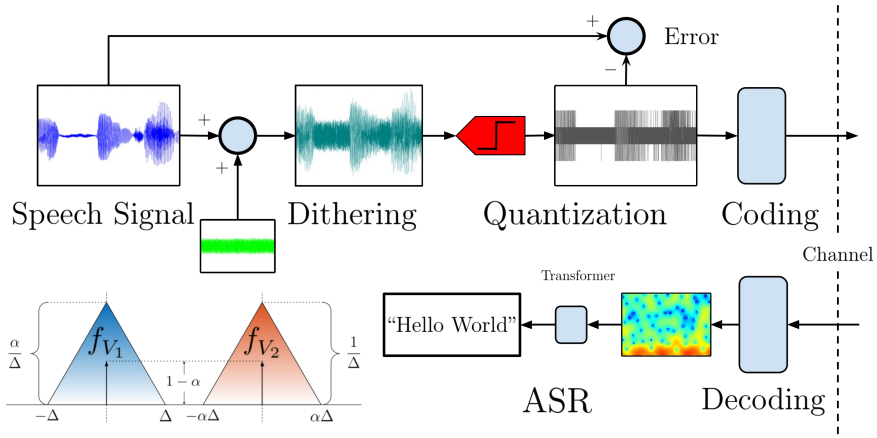
METHOD

- We design two dithers to optimizes between competing objectives:
  - Minimize **harmonic distortion**.
  - Minimize the increase in **additive noise power**.

$$\mathrm{CER}(\alpha) \propto \mathrm{MSE}(\alpha) + \mathrm{ACF}_5(\alpha)$$

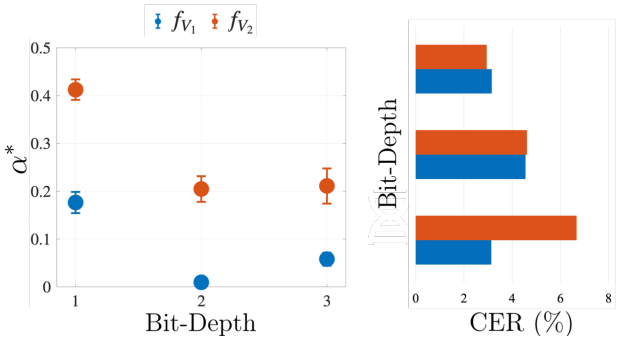
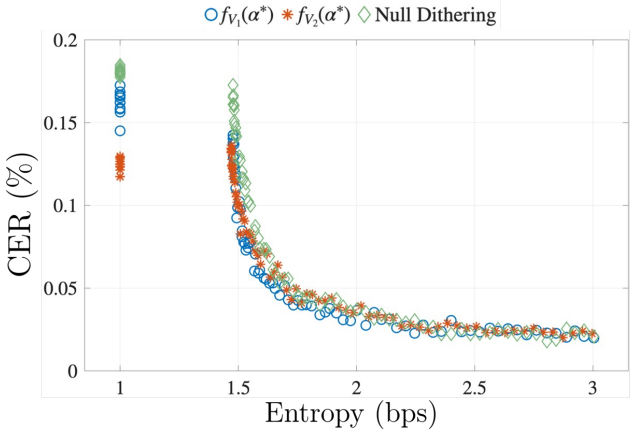
- **Choice 1 ($f_{V_1}(\alpha)$):** controls the sparsity of the dither.
- **Choice 2 ($f_{V_2}(\alpha)$):** controls the support of the dither.





RESULTS

- ASR performance is evaluated by **Character Error Rate** of the final the quantized speech.
- For ASR performance **MSE** holds more weight than **ACF_5**.



- Parametric dithering shows a:
  - Stable optimal α^* values
  - Superior performance to Null and Full Dithering
  - CER improvement of 6.67% at 1-bit.
  - Minimized entropy increase.